Task 4 Related work and optional forecasting extension

IS403.R11 Group 5

Owners: Trần Hoàng Huy, Lê Minh Tiến

Stage 4 | Handoff: 1 December 2026

Explain how the three studies inform this project and decide whether SARIMA or SARIMAX is feasible.

Inputs

The supplied introductions, the three original papers listed in docs/references.md, the daily panel and any available oil-price series.

Work

1. Write one short note per paper: question, data frequency, unit, methods, findings, limitation and what this group adopts. Verify citations against the originals before quoting results.

2. Suggested split: Huy leads the comparison and AIS studies; Tiến leads the SARIMAX study and oil data audit. Review and combine the notes.

3. Explain the differences between annual or monthly TEU, weekly commodity cargo and this project's daily estimated tonnes. Do not claim to reproduce a study using a different target.

4. By the deadline, write a go or no-go decision based on coverage, frequency, publication timing, missing values, runtime and whether all core models are complete.

5. If feasible, fit per-port SARIMA on a training-selected subset, forecast seven daily values and sum them. For SARIMAX, use only oil values available at each forecast origin; future oil paths must be forecast or stated scenarios, never future observed prices.

Deliverables

File or location	Expected output
docs/related_work.md	English Related Work and adoption table
docs/extension_decision.md	Go or no-go with evidence
src/sarimax.py	Only if the extension proceeds
results/extension_predictions.csv	Only if feasible; identify the subset

Completion check

The no-go decision is a valid deliverable. If implemented, compare against core models on the same subset and origins. Monthly oil observations cannot be treated as known before their publication date.